Using ONTAP as persistent storage for Kubernetes

Cloud Manager 3.7

Ben Cammett
January 03, 2020

This PDF was generated from https://docs.netapp.com/us-en/occm37/task_connecting_kubernetes.html on October 28, 2020. Always check docs.netapp.com for the latest.

Table of Contents

Using ONTAP as persistent storage for Kubernetes 1

Using ONTAP as persistent storage for Kubernetes

Cloud Manager can automate the deployment of NetApp Trident on Kubernetes clusters so you can use ONTAP as persistent storage for containers. This works with Cloud Volumes ONTAP and on-prem ONTAP clusters.

Before you complete these steps, you need to create a Cloud Volumes ONTAP system or discover an on-premises ONTAP cluster from Cloud Manager.

If you deploy Kubernetes clusters using the NetApp Kubernetes Service, Cloud Manager can automatically discover the clusters from your NetApp Cloud Central account. If that's the case, skip the first two steps and start with step 3.

1 Verify network connectivity

a. A network connection must be available between Cloud Manager and the Kubernetes clusters, and from the Kubernetes clusters to ONTAP systems.

b. Cloud Manager needs an outbound internet connection to access the following endpoints when installing Trident:

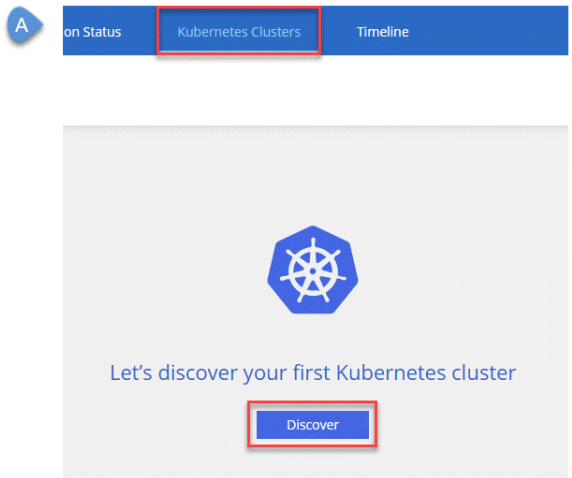
   https://packages.cloud.google.com/yum
   https://github.com/NetApp/trident/releases/download/

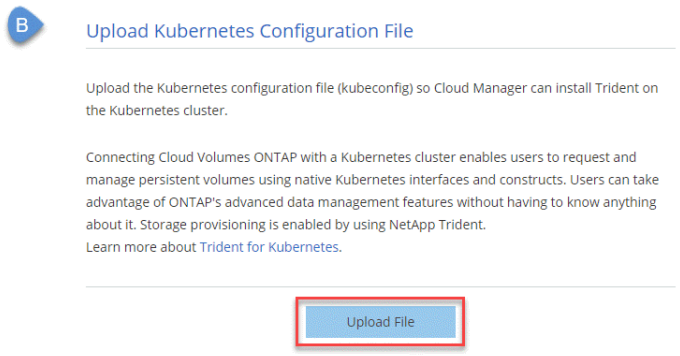
   Cloud Manager installs Trident on a Kubernetes cluster when you connect a working environment to the cluster.

2 Upload Kubernetes configuration files to Cloud Manager

For each Kubernetes cluster, the Account Admin needs to upload a configuration file (kubeconfig) that is in YAML format. After you upload the file, Cloud Manager verifies connectivity to the cluster and saves an encrypted copy of the kubeconfig file.

Click **Kubernetes Clusters** > **Discover** > **Upload File** and select the kubeconfig file.

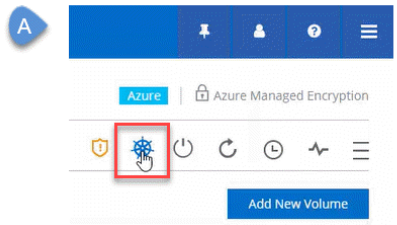


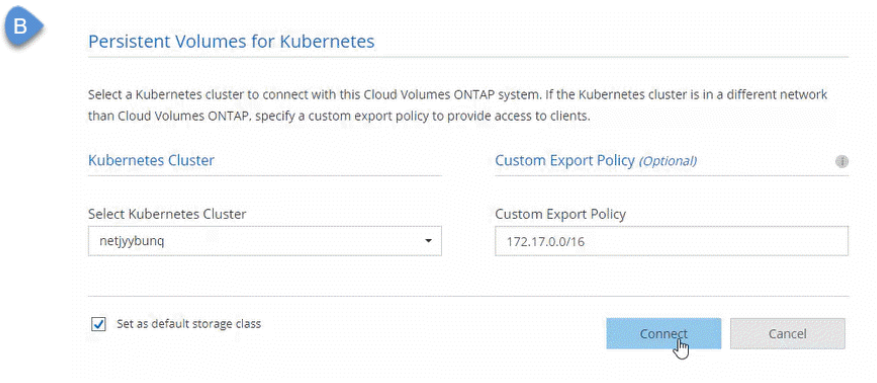


3 Connect your working environments to Kubernetes clusters

From the working environment, click the Kubernetes icon and follow the prompts. You can connect different clusters to different ONTAP systems and multiple clusters to the same ONTAP system.

You have the option to set the NetApp storage class as the default storage class for the Kubernetes cluster. When a user creates a persistent volume, the Kubernetes cluster can use connected ONTAP systems as the backend storage by default.





4 Start provisioning Persistent Volumes

Request and manage Persistent Volumes using native Kubernetes interfaces and constructs. Cloud Manager creates four Kubernetes storage classes that you can use when provisioning Persistent Volumes:

- **netapp-file**: for binding Persistent Volumes to single-node ONTAP systems
- **netapp-file-san**: for binding iSCSI Persistent Volumes to single-node ONTAP systems
- **netapp-file-redundant**: for binding Persistent Volumes to ONTAP HA pairs
- **netapp-file-redundant-san**: for binding iSCSI Persistent Volumes to ONTAP HA pairs

Cloud Manager configures Trident to use the following provisioning options by default:

- Thin volumes
- The default Snapshot policy
- Accessible Snapshot directory

Learn more about provisioning your first volume with Trident for Kubernetes

What are the trident_trident volumes?

Cloud Manager creates a volume on the first ONTAP system that you connect to a Kubernetes cluster. The name of the volume is appended with "_trident_trident." ONTAP uses this volume to connect to the Kubernetes cluster. You should not delete these volumes.

What happens when you disconnect or remove a Kubernetes cluster?

Cloud Manager enables you to disconnect individual ONTAP systems from a Kubernetes cluster. When you disconnect a system, you can no longer use that ONTAP system as persistent storage for containers. Existing Persistent Volumes are not deleted.

After you disconnect all systems from a Kubernetes cluster, you can also remove the entire Kubernetes configuration from Cloud Manager. Cloud Manager does not uninstall Trident when you remove the cluster and it does not delete any Persistent Volumes.

Both of these actions are available through APIs only. We plan to add the actions to the interface in a future release.
Click here for details about the APIs.

Copyright Information

Copyright © 2020 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.